Climate Field Reconstruction Under Station Sparsity and Geographic Shift

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Abstract

Uneven station coverage limits regional analysis of temperature, precipitation, and climate extremes. We test whether compact neural models improve same-day reconstruction from European stations and remain reliable under station sparsity and geographic shift. Using ECA&D observations and ERA5 reference fields, we compare a latent-grid transformer and ConvCNP with tuned local kriging across eight training seeds. The transformer reduces in-domain temperature RMSE from 2.21 to 1.16 °C, while precipitation RMSE is smaller, from 3.80 to 3.45 mm/day. Under eastward geographic transfer, kriging remains near 2.24 °C, whereas neural errors rise to 4.12-4.73 °C. Conformal recalibration restores coverage only through wider intervals. Neural models can improve reconstruction in familiar regions, but geographic shift must be evaluated before reconstructed fields support climate analysis.

1 Introduction and Related Work

Climate field reconstruction estimates a gridded climate variable from sparse and irregular observations. Daily gridded temperature and precipitation products are used for regional climate analysis and studies of extremes, but their accuracy depends on the station network and interpolation method [Haylock et al., 2008, Hofstra et al., 2009]. We reconstruct daily mean temperature and precipitation over Europe using same-day ECA&D station measurements, using ERA5 as a reference analysis rather than validated ground truth. This is not a forecasting problem: although the test years follow training, each reconstruction uses observations from the date being reconstructed. We compare tuned local ordinary kriging with compact ConvCNP and latent-grid transformer models. Kriging interpolates observations through a specified spatial covariance model, whereas learned methods can represent nonlinear relationships among observations and coordinates. ConvCNPs have been applied to environmental prediction from irregular spatial observations [Gordon et al., 2020, Vaughan et al., 2022], while transformer neural processes use gridded pseudo-tokens to process large, unstructured spatiotemporal datasets [Ashman et al., 2025]. We ask whether the added model complexity produces consistent improvements when the station network becomes sparse, the evaluation region changes, or conditions are extreme. We report RMSE and MAE together with the empirical coverage and width of nominal 90% conformal prediction intervals. Split conformal prediction uses held-out errors to calibrate uncertainty intervals [Vovk et al., 2005, Angelopoulos and Bates, 2023]. We test whether calibration from western Europe and the full station network remains valid after eastward transfer and station thinning. Separately, labeled eastern data measure the interval width needed to recover coverage toward 90%. Using matching inputs, targets, and temporal splits, we evaluate later years, geographic transfer, station thinning, and extreme days, averaging neural results across eight seeds.

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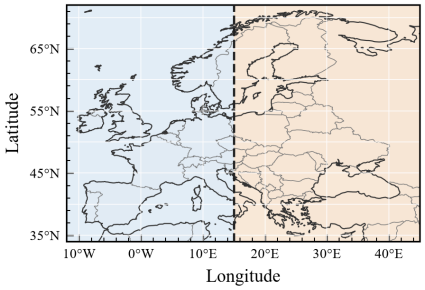


Figure 1: Geographic transfer map of Europe, showcasing the 15°E line for west to east transfer.

2 Methods and Experimental Design

Problem Formulation: For day t, let $\mathcal{C}_t = \{(\mathbf{s}_i, y_i)\}_{i=1}^{n_t}$ be the available station locations and same-day measurements. Given $\mathcal{C}_t$ and target locations $\mathcal{G}_t \subseteq \mathcal{G}$, each method predicts a mean $\hat{\mu}_t(\mathbf{g})$ and uncertainty scale $\hat{\sigma}_t(\mathbf{g})$ for each $\mathbf{g} \in \mathcal{G}_t$. These outputs are used to construct conformal prediction intervals. Temperature is modeled in degrees Celsius. Precipitation is modeled in standardized $\log(1+y)$ space; RMSE and MAE use the inverse-transformed median $\max\{\exp(\hat{\mu}_{\log}) - 1, 0\}$, and interval endpoints are transformed back to mm/day.

Reconstruction methods:
Ordinary kriging predicts each target from a local neighborhood of stations using an exponential covariance and a nugget equal to 5% of the daily context variance [Cressie, 2015]. The neighborhood size and length scale are selected by mean daily RMSE on 30 validation days.
ConvCNP maps irregular station observations onto a 64×64 latent grid using Gaussian distance weights, applies two convolutional hidden layers, and queries the resulting field at arbitrary target coordinates [Gordon et al., 2020, Vaughan et al., 2022].
The **latent-grid transformer** uses a simplified gridded pseudo-token architecture [Ashman et al., 2025]. A fixed 16×16 grid attends to embedded station values and coordinates, and target location queries attend to the resulting latent representation. Both neural models output a Gaussian mean and log standard deviation and are trained using Gaussian negative log likelihood.

For each run, we evaluate the checkpoint with the lowest validation negative log likelihood; full training and stopping details are provided in Appendix A.

3 Experiments

Data: We use ECA&D daily mean temperature (`TG`) and daily precipitation totals (`RR`) [Klein Tank et al., 2002]. After restricting stations to the ERA5 domain, the data contains 6,500 temperature and 10,039 precipitation station identifiers, with availability varying by day (Appendix B). Targets are daily ERA5 fields at $0.25°$ resolution [Hersbach et al., 2020], expressed in °C and mm/day. ERA5 is treated as a reference analysis rather than error-free ground truth. We use 2010-2017 for training, 2018 for validation and model selection, 2019 for conformal calibration, and 2020-2021 for testing.

Evaluation settings:
Full domain. Model fitting, calibration, and evaluation use the full European domain. For **geographic transfer**, models are trained using observations and calibrated west of or at 15°E and evaluated east of 15°E. For each eastern test date, the models use that day's eastern station measurements as inputs, but their parameters were learned only from western-region data. For **extreme events**, we evaluate the same models without event-specific retraining. Following the general use of percentile-based climate extreme indices [Zhang et al., 2011], we identify events from ERA5 daily spatial mean anomalies relative to a 31-day climatology estimated from 2010-2017. Temperature extremes fall below the training 5th percentile or above the 95th percentile, while heavy precipitation days exceed the training 95th percentile. Because station network density can affect interpolation accuracy [Hofstra et al., 2009], **station sparsity evaluation** retains 100%, 75%, 50%, 25%, and 10% of the stations available on each test day. The same retained stations are used for every method, so comparisons are not affected by differences in the sampled station network. See Appendix C for station selection details.

Table 1: Full-domain extreme event performance. Coverage is the empirical coverage of nominal 90% conformal intervals, neural methods averaged over eight seeds.

	Precipitation (mm/day)			Temperature (°C)		
Model	RMSE	Cov. (%)	Width	RMSE	Cov. (%)	Width
Kriging	4.93	91.5	22.48	2.34	87.6	6.82
ConvCNP	4.65	88.5	7.93	1.52	91.7	4.68
Latent-grid Transformer	4.58	87.1	6.99	1.22	90.2	3.54

Evaluation metrics: Point accuracy is measured using RMSE and MAE in physical units, computed separately for each test day and then averaged across days. For nominal 90% conformal prediction intervals, we report day-averaged empirical coverage,

$$\widehat{\mathrm{Cov}} = \frac{1}{T}\sum_{t=1}^{T}\frac{1}{n_t}\sum_{j=1}^{n_t}\mathbf{1}\{y_{tj} \in [L_{tj}, U_{tj}]\}, \tag{1}$$

where T is the number of test days, n_t is the number of sampled grid cells on day t, and y_{tj} is the ERA5 reference value. We report mean conformal interval width alongside coverage because wider intervals can achieve high coverage while providing less precise reconstructions. For each neural model, metrics are computed separately for eight independently trained seeds. Values written as $\pm$ report the mean and standard deviation across seeds, while figure bands show 95% confidence intervals. Kriging is deterministic and is reported without a seed-based interval.

4 Results

At full station density, the latent-grid transformer has the lowest full-domain RMSE for temperature (1.16 °C, compared with 1.53 for ConvCNP and 2.21 for kriging) and precipitation (3.45 mm/day, compared with 3.51 and 3.80, respectively).

On extreme days, shown in Table 1, both neural models reduce RMSE relative to kriging, with the largest improvement occurring for temperature. The latent-grid transformer has the lowest temperature RMSE and maintains 90.2% coverage with the smallest interval width. Precipitation RMSE decreases marginally between methods, however, the neural precipitation intervals are substantially narrower, their coverage falls to 88.5% for ConvCNP and 87.1% for the latent-grid transformer, compared with 91.5% for kriging. Thus, the learned models improve accuracy on extreme days, especially for temperature, but do not provide a consistent uncertainty advantage for precipitation.

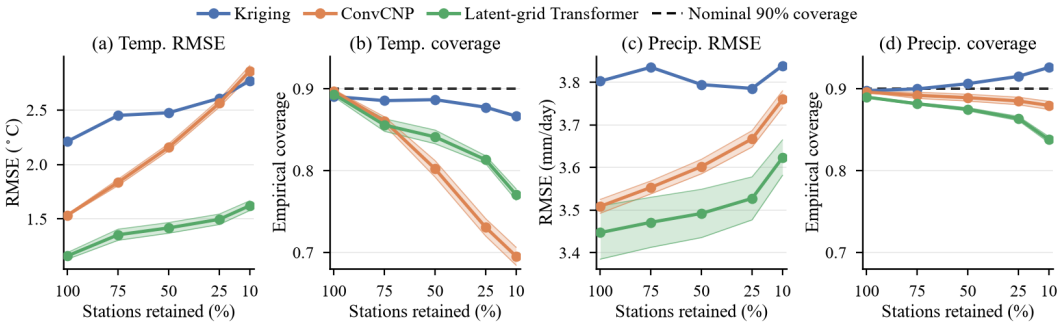


Figure 2: RMSE and nominal 90% conformal coverage under station sparsity tests. Neural curves show eight-seed means with 95% confidence bands. Intervals are calibrated at 100% retention and not recalibrated after thinning.

Figure 2 shows how reducing station coverage from 100% to 10% affects temperature more drastically than precipitation. Temperature RMSE increases for every method as fewer stations are retained, although latent-grid transformer maintains the lowest RMSE across the full retention range. ConvCNP remains slightly more accurate than kriging down to 25% retention but becomes slightly less accurate at 10%. The neural intervals also lose coverage throughout: temperature coverage falls from 89.7% to 69.5% for ConvCNP and from 89.3% to 77.1% for the latent-grid transformer, while kriging changes

from 89.0% to 86.7%. Precipitation RMSE changes less, but neural coverage still declines as stations are removed.

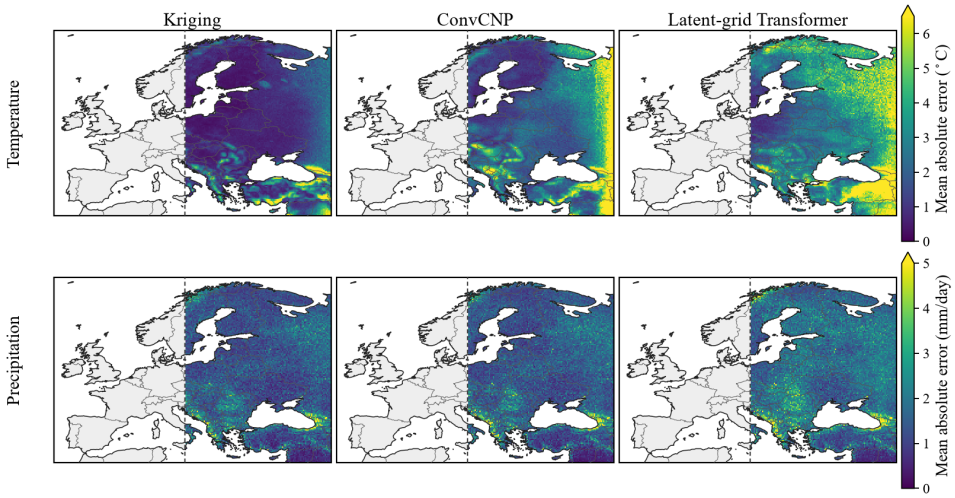


Figure 3: Mean absolute error under eastward transfer for a single neural seed. Temperature is capped at the 95th percentile of plotted errors and precipitation at 5 mm/day. The dashed line marks $15°$E.

Across eight neural seeds, temperature RMSE increases to $4.12 \pm 0.20°$C for ConvCNP and $4.72 \pm 0.32°$C for the latent-grid transformer, compared with $2.24°$C for kriging. For precipitation, kriging obtains an RMSE of 3.19 mm/day and the latent-grid transformer obtains 3.78 ± 0.11 mm/day. ConvCNP reaches 22.44 ± 52.19 mm/day, with the large standard deviation indicating strong variation with the training seed; additional data found in Appendix D).

Figure 3 illustrates transfer error for one neural seed. Kriging has lower temperature error across much of the eastern region, while precipitation differences are more localized. Because the neural maps show one seed, the aggregate metrics provide the main evidence about geographic transfer.

5 Discussion and Limitations

Model complexity helps in-domain but not consistently under distribution shift. Both neural models reduce RMSE on extreme days, and the latent-grid transformer remains the most accurate under station sparsity, although ConvCNP falls behind kriging at 10% station retention. Under transfer, temperature RMSE for both neural models is approximately twice that of kriging. Precipitation is mixed, with ConvCNP showing strong variation across training seeds. Thus, the evaluated neural models do not transfer as consistently as local kriging.

These conclusions are limited by the $15°$E split. ERA5 is used as the reconstruction target, and any spatial bias it carries is inherited by all three methods. Our findings apply to the ConvCNP and latent-grid transformer implementations, and should not be generalized to all learned reconstruction methods. Conformal calibration reflects only the average coverage across the full test region and period, which can mask trends in local or seasonal spaces. The geoplots only contain one seed of training and should be interpreted as illustrative rather than aggregate evidence.

Accurate reconstruction methods are essential for understanding and responding to climate change, especially in regions where weather station coverage is sparse. Improving the estimates of climate variables at unobserved locations could benefit applications such as agricultural planning and climate risk assessment. Our results establish that neural approaches can substantially improve reconstruction accuracy under certain conditions, while Kriging is more reliable under geographic transfer. Understanding these tradeoffs can help researchers select appropriate methods for each environment.

Furthermore, model selection also raises questions about climate justice, especially when models are trained in regions with plentiful weather station coverage but deployed elsewhere. Neural networks trained on data from regions with plentiful observations such as Western Europe, may overfit to regional climate conditions, and struggle to generalize to regions with different conditions such as

Eastern Europe. Simply evaluating models on regions similar to their training data can misrepresent how reliable they are, potentially disadvantaging communities that are already affected by limited climate infrastructure. Addressing these concerns requires careful consideration of data diversity and model robustness, so that advances in climate modeling can benefit all regions equally.

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A Optimization and Implementation Details

Neural architectures. The ConvCNP uses a 64×64 grid, 64 hidden channels, and Gaussian kernel projection with $\sigma = 0.08$ in normalized coordinate units. Station values are embedded by a two-layer MLP. The values and station density channel pass through two 5×5 convolutional layers followed by a 1×1 Gaussian output head. The latent-grid transformer has embedding dimension 128, four attention heads, three latent self-attention layers, feed-forward dimension 512, dropout 0.1, and a 16×16 latent grid. Coordinates are encoded using their normalized values and six sinusoidal frequencies. Both models output a Gaussian mean and log standard deviation, with log standard deviation restricted to $[-7, 5]$.

Optimization. Coordinates are scaled to $[-1, 1]$, and climate values are standardized using statistics from the training period station observations in the corresponding training region. Precipitation is transformed using $\log(1 + y)$ before standardization. Each training batch contains one day, all available stations in the training region, and 1,024 ERA5 grid cells sampled without replacement. Training targets are resampled each epoch, while validation targets remain fixed. We use AdamW [Loshchilov and Hutter, 2019] with learning rate 3×10^{-4}, weight decay 10^{-4}, and gradient-norm clipping at 1.0, implemented in PyTorch [Paszke et al., 2019]. No learning-rate schedule is used. For precipitation, we impose a minimum predictive standard deviation of 0.1 in normalized log-transformed units.

Checkpoint selection and stopping. Models are trained for at most 50 epochs and at least 10 epochs. Validation negative log likelihood is evaluated after every epoch. Training stops after eight consecutive epochs without improvement, and evaluation uses the checkpoint with the lowest validation loss. This stopping rule is our operational convergence criterion. We repeat training with seeds 2026-2033. Table 2 reports the selected checkpoint and stopping epoch ranges across the eight seeds.

Table 2: Selected-checkpoint and stopping epochs across eight training seeds. "Full" is training over the full European domain and "West" is training west of or at $15°$E.

Variable	Model	Domain	Best epoch	Stop epoch
Temperature	ConvCNP	Full	13–37	21–45
Temperature	ConvCNP	West	11–44	19–50
Temperature	Latent-grid Transformer	Full	30–50	38–50
Temperature	Latent-grid Transformer	West	18–48	26–50
Precipitation	ConvCNP	Full	25–46	33–50
Precipitation	ConvCNP	West	13–48	21–50
Precipitation	Latent-grid Transformer	Full	19–49	27–50
Precipitation	Latent-grid Transformer	West	21–36	29–44

Kriging tuning. The kriging neighborhood size is selected from $\{16, 32\}$ stations using mean daily RMSE on 30 dates from the 2018 validation split. Temperature length scales are selected from $\{200, 400, 800\}$ km, giving a final setting of 32 stations and 800 km. Precipitation additionally considers 100 km and selects 32 stations and 200 km. Both variables use an exponential covariance and a nugget equal to 5% of the daily context variance.

Computational resources. Latent-grid transformer runs use an NVIDIA GeForce RTX 4080 SUPER with 16 GB of memory, while ConvCNP runs use an NVIDIA GeForce RTX 5060 Ti with 16 GB. Kriging runs on CPU. Neural training uses PyTorch 2.11.0 with CUDA 12.8. The 64 neural training runs, covering two variables, two models, two training regions, and eight seeds, total approximately 26.0 recorded GPU-hours.

B Data processing and temporal splits

Station processing. We parse the ECA&D blended station files for daily mean temperature (`TG`) and precipitation total (`RR`) [Klein Tank et al., 2002]. Records are retained when the ECA&D quality flag is zero and the value is not missing; negative precipitation records are also removed. Station files are included when more than 90% of their available records during 2010-2021 pass these checks. This threshold measures the quality of reported records, not completeness across all calendar days. Raw

TG and RR values are divided by ten to obtain °C and mm/day. We then remove rows with missing coordinates or values and restrict stations to the ERA5 domain, $25°$W-$45°$E and $34°$N-$72°$N. The resulting temperature table contains 6,500 station identifiers and 26,945,775 station-day observations, with 5,735-6,433 observations per day (median 6,163). The precipitation table contains 10,039 station identifiers and 43,062,177 station-day observations, with 9,485-9,994 observations per day (median 9,836). All methods receive the same available station observations on each date.

ERA5 processing. ERA5 reference fields use a 153×281 grid at $0.25°$ resolution, giving 42,993 grid cells per day [Hersbach et al., 2020]. Hourly 2 m temperature fields are averaged by UTC day and converted from kelvin to °C during model loading. Daily precipitation is the sum of 24 hourly total-precipitation accumulations, converted from metres to mm/day; negative accumulated values are set to zero. Both reference datasets contain 4,383 daily fields from 1 January 2010 through 31 December 2021.

Temporal splits and normalization. The chronological splits contain 2,922 training days (from 2010-2017), 365 validation days (from 2018), 365 conformal-calibration days (from 2019), and 731 test days (from 2020-2021). Each date appears in a single split. We standardized Temperature values using the mean and standard deviation of station observations from the model's training region during the training period. Before applying the same standardization to precipitation, we transformed it using $\log(1 + y)$. Coordinates are linearly rescaled using the fixed ERA5 domain bounds.

C Evaluation Protocol Details

The geographic boundary is fixed at $15°$E. The western source region contains locations with longitude at or below the boundary, while the eastern target region contains locations above it. Table 3 summarizes the three evaluation settings. Target-region recalibration changes only the conformal correction and interval width; it does not change the trained model or its point predictions.

Table 3: Training, calibration, and test regions for each evaluation setting.

Setting	Model fitting	Calibration	Evaluation
Full domain	Full domain	Full domain	Full domain
Source-calibrated transfer	West	West	East
Target-recalibrated transfer	West	East	East

For both transfer settings, the source-trained model receives eastern station measurements from the date being reconstructed. Target-region recalibration uses labeled eastern data from 2019 and is included as a diagnostic of the interval width required after transfer.

C.1 Station retention

Station thinning uses a deterministic pseudo-random ordering shared by every method and neural seed. For station identifier i, we compute

$$r_i = (2654435761\, i + s) \bmod 2^{32}, \tag{2}$$

where s is a fixed evaluation seed. On a day with n_t available stations, retention level p keeps the first

$$k_t(p) = \max\left(1, \text{round}(pn_t)\right) \tag{3}$$

available stations after sorting by r_i. We evaluate $p \in \{1.00, 0.75, 0.50, 0.25, 0.10\}$.

Because every retention level uses a prefix of the same ordering, the retained networks are nested: stations present at 10% retention are also present at 25%, 50%, 75%, and 100%. Availability may differ by day, but each station keeps the same rank whenever it is available. Temperature uses evaluation seed 9102 and precipitation uses seed 19102. The same retained stations are supplied to kriging, ConvCNP, and the latent-grid transformer. This design isolates the effect of station count from variation caused by independently redrawing a network for each method or retention level.

For the station-thinning results in Figure 2, conformal intervals are calibrated at 100% retention and applied without recalibration at lower retention levels. The resulting coverage therefore measures whether full-network calibration remains valid after stations are removed.

C.2 Extreme-event selection

Extreme days are selected independently for each evaluation region using ERA5. For each date, we first compute the unweighted spatial mean over ERA5 grid cells in that region. A calendar-day climatology is then estimated from 2010–2017 using all training dates within 15 days of each calendar day, with circular wrapping at the start and end of the year. This produces a 31-day climatological window. Daily anomalies are calculated by subtracting this climatology from the corresponding regional mean.

Temperature thresholds are the 5th and 95th percentiles of the training-period anomaly distribution. Test dates are classified as extreme when their anomaly is at or below the lower threshold or at or above the upper threshold. Heavy-precipitation dates are those at or above the training-period 95th percentile. The models are neither retrained nor recalibrated on the selected event days.

Table 4: Training-period anomaly thresholds and numbers of selected extreme days in the 731-day test period. Temperature thresholds are in °C and precipitation thresholds are in mm/day.

Variable	Region	Lower threshold	Upper threshold	Selected days
Temperature	Full domain	-1.95	1.53	106
Temperature	East	-3.49	3.03	104
Precipitation	Full domain	n/a	0.82	41
Precipitation	East	n/a	1.24	36

C.3 Evaluation sampling and conformal calibration

Each evaluation day uses 1,024 ERA5 target cells sampled without replacement. The target samples and station subsets are fixed across methods. Temperature uses calibration seed 9101 and evaluation seed 9102; precipitation uses calibration seed 19101 and evaluation seed 19102.

For calibration target j, the normalized nonconformity score is

$$a_j = \frac{|y_j - \hat{\mu}_j|}{\max(\hat{\sigma}_j, 10^{-6})}. \tag{4}$$

For n calibration scores and nominal coverage $1 - \alpha = 0.90$, we select the ordered score at rank

$$k = \lceil (n+1)(1-\alpha) \rceil \tag{5}$$

and construct

$$[\hat{\mu} - q\hat{\sigma},\ \hat{\mu} + q\hat{\sigma}], \tag{6}$$

where q is the selected score. Precipitation intervals are constructed in log-transformed space and converted back to mm/day. Calibration uses all 2019 dates rather than only extreme-event dates.

D Additional Results

Table 5: Complete reconstruction and uncertainty results. Values are reported as mean ± standard deviation over eight seeds.

	Precipitation (mm/day)				Temperature (°C)			
Model	RMSE	MAE	Cov. (%)	Width	RMSE	MAE	Cov. (%)	Width
Global evaluation								
Kriging	3.80	1.83	89.8	11.97	2.21	1.53	89.0	6.96
ConvCNP	3.51 ± 0.02	1.69 ± 0.01	89.6 ± 0.1	6.09 ± 0.20	1.53 ± 0.01	1.12 ± 0.01	89.7 ± 0.2	4.60 ± 0.06
Latent-grid Transformer	3.45 ± 0.07	1.64 ± 0.02	89.0 ± 0.1	5.71 ± 0.22	1.16 ± 0.04	0.82 ± 0.03	89.3 ± 0.4	3.36 ± 0.10
Geographic transfer with transferred calibration								
Kriging	3.19	1.44	89.4	6.97	2.24	1.48	87.0	6.15
ConvCNP	22.44 ± 52.19	3.37 ± 5.07	89.0 ± 0.7	104.75 ± 160.82	4.12 ± 0.20	2.85 ± 0.07	62.9 ± 1.8	10.42 ± 11.81
Latent-grid Transformer	3.78 ± 0.11	1.88 ± 0.09	80.2 ± 2.9	5.05 ± 0.97	4.72 ± 0.32	3.61 ± 0.34	43.0 ± 4.8	4.90 ± 0.92
Geographic transfer with target-region recalibration								
Kriging	3.19	1.44	89.7	7.21	2.24	1.48	88.4	6.56
ConvCNP	22.44 ± 52.19	3.37 ± 5.07	89.4 ± 0.1	105.15 ± 160.58	4.12 ± 0.20	2.85 ± 0.07	89.2 ± 0.1	25.59 ± 31.64
Latent-grid Transformer	3.78 ± 0.11	1.88 ± 0.09	89.2 ± 0.4	10.32 ± 3.25	4.72 ± 0.32	3.61 ± 0.34	89.2 ± 0.4	16.19 ± 1.79

This table presents the complete reconstruction and uncertainty results for the global and transfer experiments. For the neural models, each value is reported as the mean and standard deviation across eight training seeds, while Kriging produced a single result. Reconstruction performance is evaluated using RMSE and MAE. Confidence and uncertainty is evaluated using empirical coverage and conformal interval width.